

HE LI

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EDUCATION

Tsinghua University (THU)

July 2026 (expected)

B.E. in Computer Science (Yao Class, IIIS)

GPA: 3.955/4.000

Rank: 1/94 among Yao Class

RESEARCH INTERESTS

I am interested in modern visual generative model and multimodal language model, together with the vision-centric applications in the other domains.

RESEARCH EXPERIENCES

3D Gaussian Splatting

Oct 2025 - Present

Advisors: Prof. Yiming Li

THU (Undergraduate Thesis)

- Exploring 3D Gaussian Splatting in visual encoder's latent space. (In processing)

Multimodal LLM and AI for Biology

Jun 2025 - Sep 2025

Advisors: Prof. Yeung-Levy Serena

Stanford, Onsite UGVRI

- Leading project on analyzing Multimodal LLM natural robustness against catastrophic forgetting [1] and its potential application in medical foundation model [2].
- Participating in CellFlux V2 project aiming for simulating cellular morphology changes via Flow Matching. Accelerating the training by 6 times and developing the results on JUMP datasets [3].

Visual Generative Model

Feb 2024 - May 2025

Advisors: Prof. He Kaiming

MIT, Onsite RA in Spring 2025

- Exploring the autoregressive method without vector quantisation [4]. Leading project on improving this methods with adversarial training, one-step diffusion-based method and causal autoregressive.
- Leading project on finetuning generation model from pretrained representation model. Also investigating the co-training of representation models and generative models.
- Help developing a new family of single-step generative model. (In processing, with Prof. Du Yilun)

Sparsity for Diffusion Models

Oct 2023 - Jan 2024

Advisors: Prof. Chen Jianfei and Prof. Zhu Jun

THU (Undergrad. Intern)

- Finish baseline testing and help develop progressive $N : M$ sparsity method for diffusion model [5].

AWARDS & GRANTS

Yao Award

2025

Technological Innovation Scholarship

2025

Widjaja Scholarship

2025

Academic Excellence Scholarship

2024

Nanjing Turing Institute of Artificial Intelligence Scholarship

2024

Tsinghua Freshman Scholarship

2022-2026

First prize in provincial CMO (Tianjin)

2020, 2021

First prize in provincial CPhO (Tianjin)

2020, 2021

First prize in CSP-S (Tianjin)

2019

ACADEMIA & TEACHING SERVICES

Reviewer for ICCV, ARR, ML4H, NeurIPS, CVPR	2025
TA for Machine Learning by Prof. Yuan Yang	Sep 2025 - Jan 2026
TA for Advanced Computer Graphics by Prof. Yi Li	Sep 2025 - Jan 2026
Student TA for Object-Oriented Programming by Prof. Liu Zhiyuan	Feb 2023 - July 2023

PUBLICATIONS

- [1] **He Li**, Y. Zhang, X. Wang, K. Lyu, and S. Yeung-Levy, Fine-tuning vlms without forgetting is easier than you think, in *ResponsibleFM @ NeurIPS 2025, ICLR 2026 Submission*.
- [2] **He Li**, Y. Zhang, X. Wang, and S. Yeung-Levy, Replay-free sequential fine-tuning of medical vlms, in *ML4H 2025*.
- [3] Y. Zhang, Y. Su, Z. Wefers, S. Su, **He Li**, T. Li, C. Wang, J. Burgess, A. Lozano, L. Zhou, D. Ding, J. J. Nirschl, E. Lundberg, and S. Yeung-Levy, Cellfluxv2: An image generative foundation model for virtual cell modeling, in *ML4H 2025, Nature Method 2025 Submission*.
- [4] T. Li, Y. Tian, **He Li**, M. Deng, and K. He, Autoregressive image generation without vector quantization, in *NeurIPS 2024 Spotlight*.
- [5] K. Wang, J. Chen, **He Li**, Z. Mi, and J. Zhu, Sparsedm: Toward sparse efficient diffusion models, in *ICME 2025*.

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- [1] **He Li**, Y. Zhang, X. Wang, K. Lyu, and S. Yeung-Levy, Fine-tuning vlms without forgetting is easier than you think, in *ResponsibleFM @ NeurIPS 2025, ICLR 2026 Submission*.
- [2] **He Li**, Y. Zhang, X. Wang, and S. Yeung-Levy, Replay-free sequential fine-tuning of medical vlms, in *ML4H 2025*.
- [3] Y. Zhang, Y. Su, Z. Wefers, S. Su, **He Li**, T. Li, C. Wang, J. Burgess, A. Lozano, L. Zhou, D. Ding, J. J. Nirschl, E. Lundberg, and S. Yeung-Levy, Cellfluxv2: An image generative foundation model for virtual cell modeling, in *ML4H 2025, Nature Method 2025 Submission*.
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SELECTED OPEN-SOURCE PROJECTS

Imitation Learning with Diffusion Policy Sep 2024 - Jan 2025
Repository: Imitation Learning with Diffusion Policy Collaborators: Rujia Yang

- Incorporating Low-Dimensional Self-Supervised Loss for Diffusion Policies in Imitation Learning.

Merged Contribution to Maniskill Repository Sep 2024 - Jan 2025
Repository: Enhance SAC with MoE and BEE Operator Collaborators: Guowei Xu, Muhan Wang

- Introducing two plug-and-play enhancements to the Soft Actor-Critic (SAC) algorithm.

Physically Based GPU Graphics Renderer Sep 2024 - Jan 2025
Repository: GPU Rendering for Interference and Dispersion Collaborators: Chenglin Liu

- GPU-based graphics renderer implemented in GLSL with original wave effect simulation feature.

KAN in Computer Vision Feb 2024 - July 2024
Repository: Computer Vision Meets KAN Collaborators: Yue Cao

- Classification by FFT/PCA preprocessing and Kolmogorov-Arnold Network, achieving higher accuracy.